Softmax Activation

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Normally when we train a neural network, a forward pass yields a certain value for each class. See example below

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[[-28.75569952 -10.08059127  -6.44380494  -5.24652427
-22.95078438
  -30.35618665 -61.42638986  31.7498354  -16.79923747
0.19532748]]
```

this is a prediction made by the model where each value at an index indicates a score. this one is the result of a prediction of an mnist digit classifier, and since the largest score is at index 7, the model "thinks" the number it's seeing is 7.

Now this works, but since each score can be any value, it is hard to interpret the results. which is not cool.

Because of this; we use the softmax activation function, which converts each score into a value between 0 and 1. Essentially converting each score into a probability.

Softmax function formula:

Let $A = \{P_1, P_2 \ldots P_n\}$ → the list containing all the prediction scores.

Then, for a prediction P_a, $\text{Softmax}(P_a) =$

$$\frac{e^{P_a}}{\sum_{i=1}^{n} e^{P_i}}$$

Softmax forward pass:

we can use softmax for training. The idea behind it is that if we increase the "probability" of the correct prediction, the "probabilities" of the wrong predictions will shrink.

So it is enough to calculate the loss of the correct prediction.

Since the truth label look like $[0, \ldots 1, \ldots]$, a simple idea would be:

$(1 - \text{correct_prediction})$

This works, but to train more effectively, we will use

$-\log(\text{correct_prediction})$

which punishes confidence in wrong answers

<u>Softmax backward pass:</u>

Let $A =$ → list containing every raw output

Let $Z = \sum_{i=1}^{n} e^i$

Let $c =$ Raw output of correct prediction

Let $P_c = e^c / Z$ → correct probability

Let $L = -\log(P_c)$ → the loss

we want dL/dz_j for every j

$dL/dA = dL/dP_c \cdot dP_c/dz_j$

- $dL/dP_c = -\frac{1}{P_c}$

- dP_c/dz_j

correct case: $j = c$

$$P_c = \frac{e^c}{Z} \Rightarrow dP_c/dz_c \left(\frac{e^c}{Z}\right) = \frac{Z \cdot dP_c/dz_c\,(e^c) - e^c \cdot dP_c/dz_c\,(Z)}{Z^2}$$

$$\frac{e^c \cdot Z - e^c \cdot e^c}{Z^2} = P_c(1 - P_c)$$

wrong case:

$$dP_c/dz_c \left(\frac{e^c}{Z}\right) = \frac{Z \cdot dP_c/dz_j\,(e^c) - e^c \cdot dP_c/dz_j\,(Z)}{Z^2} = \frac{0 - e^c \cdot e^j}{Z^2} = -P_c P_j$$

- $dL/dz_c = -\frac{1}{P_c} \cdot P_c(1 - P_c) = P_c - 1$
- $dL/dz_j = -\frac{1}{P_c} \cdot -P_c P_j = P_j$

Since for the list of actual Y, only $Y_c = 1$, and $Y_j = 0$, we can generalize dL/dz_j as:

$P_j - Y_j$